A Robust Algorithm for Go Image Recognition in Go Game

Jie Song[1], Shanshan Li[1*]
[1]School of Computer Science and Technology
Anhui University, Hefei, China
*Corresponding author, e-mail: 751722205@qq.com

***Abstract*—In this paper, image of go game recognition is proposed based on Hough transform algorithm. Firstly, preprocess the go image to reduce the amount of noise. Secondly, locate four corner coordinates by Hough transform to detect lines of unoccupied board for the opening of the game. Next, most significant difference from previous work is our use of simple geometric transformation model and in process of finding the grid. Another significant improvement is the use of Hough transform for circle around intersection to locate the piece. Finally, for each intersection, compute average brightness values around it and K-Means algorithm can be determined to quantize the colors. Experimental result in abundant go images shows that the proposed method has a good recognition performance on go images of different views, go boards and a certain degree of illumination nonuniformity.**

Keywords-go image recognition; geometric transformation; Hough transform; color space

I. INTRODUCTION

The go game is a kind of Chinese chess, which has a long history and many enthusiasts all over the world, and world competitions are held every year. Figure 1 is an image of a go game. Traditionally, two players place white and black stones on the intersections of a rectangular go board, which contains of 19 × 19 lines.

Recording each move during go game is a complex manual task. It is hard to correct any mistakes. Nowadays, with the image processing and computer vision techniques available, more and more researches have been made in automating the recording each move in human go games. What's more, go image recognition is the vital basis of building go-playing robot.



Figure 1. An image of a go game.

II. RELATED WORK

The process of go images recognition can be basically divided into two major steps, board recognition and piece recognition. Board recognition refers to detection the grid of 19 × 19 lines and 361 intersections. Piece recognition means the location and classification of those pieces of intersections. Many methods have been used to recognize go images. RSC algorithm is applied to recognizing pieces on a board [1]. For board recognition, the most frequently used method is Hough transform combined with other method to detect the grid of 19 × 19 lines. Geometry-Based Populated Chessboard Recognition [2] recognizes all lines based on Hough transform, k-means clustering, and KNN applied to eliminate the outlier lines. Detecting all lines of board is based on FAPL module [3]. The Hough transform is combined with line filter in [4], [5] and mixed RANSAC in [6]. Hough transform and parameters are used to find chessboard [7]. A genetic algorithm is presented to find the contour of the go-board in [8], but do not address with all line of go board. Another approach of go board recognition is that user must manually select four corner of the board (clockwise from the top left) and then calculate a projective transformation in [9]. A machine learning method based on SIFT keypoint is used in [10]. For piece recognition, color space is a straightforward method to identify piece. Calculate average brightness around intersection in [5] and [11]. A* algorithm is applied to search for the move of piece in [6].

III. GO GAME IMAGE RECOGNITION

In this chapter, the solution of go game image recognition is described. The whole process of solution includes: preprocess, board recognition, piece recognition. Figure 2 is the flow of go game image recognition.

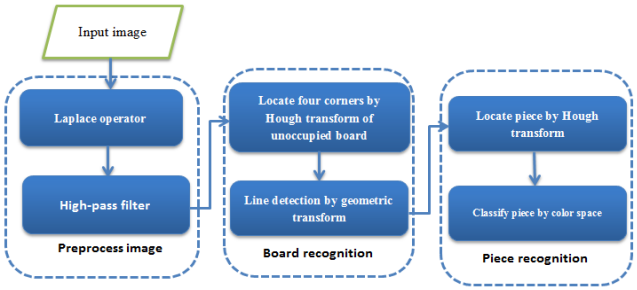


Figure 2. The flow of go game image recognition.

978-1-5386-8339-2/18/$31.00 ©2018 IEEE

Authorized licensed use limited to: Georgia Institute of Technology. Downloaded on September 27,2026 at 21:56:59 UTC from IEEE Xplore. Restrictions apply.

A. Preprocess Go Game Image

First, the image is transformed into gray-scale. Then edge detection is performed by Laplace operator to reduce noise, followed by a high-pass filter. Figure 3 is the process of processing

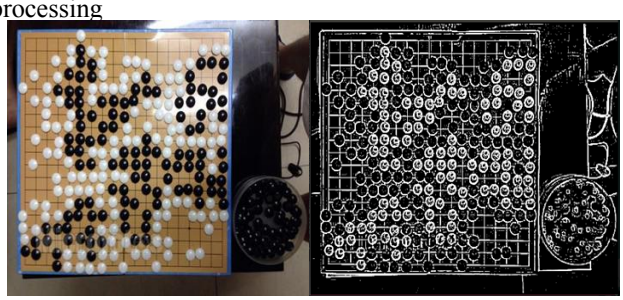


Figure 3. The process of preprocessing. The left is an input image, the right is the image after preprocessing.

B. Board Recognition

The previous board recognition is implemented using line-based methods, such as RANSAC, Hough transform or the combination of Hough transformation and other methods (RANSAC, line filter). However, by the end of a go game, massive pieces are placed on the board, and the grid lines will be covered more and more. The above methods are no longer applicable, which greatly reduces the recognition rate of grid lines and thus reduces the correct rate of piece recognition. User must manually select four corners (clockwise from the top left) in [9]. Obviously, user interaction is a time-consuming process.

Based on the above problems, the implementation of the method of this paper is as follows. Considering a go game scene, the authors assume that the board and the camera do not move during the game, so the coordinates of four corners of grid are fixed during the entire game. Since the board is always unoccupied at the start of the game, it is better for beginning with lines. The first step is that locate four corners (clockwise from the top left) by Hough transform to detect lines of unoccupied board for the opening of the game. Next, simple geometric transformation model is used to find the grid and 361 intersections.

Hough transform to detect lines was proposed by Duba and Hart [12] in 1972. A line in the image corresponds to a point in the Hough space. Each point (x, y) in Hough space is followed by two parameters as (1). For parameters bins (ρ, θ), Hough transform algorithm determine a line by three steps: first, calculate parameters values and accumulate bins; next, find local maxima which identify line; last step: peak position in Hough space represent distinct line in the image.

$$\rho = x\cos\theta + y\sin\theta \tag{1}$$

In light of the characteristics of go board: lines are parallel in the same direction and the lines of other direction are perpendicular to other half. Simple Geometric Transform Model (SGTM) is used to accurately identify the grid for go game images. As we all know, given two points (x_1, y_1), (x_2, y_2), and the equation of line is described as (2).

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1} \tag{2}$$

The equation of intersection between two lines ($a_1x+b_1y=c_1$, $a_2x+b_2y=c_2$) is described as (3).

$$(x,y) = \left(\frac{b_1c_2-b_2c_1}{a_2b_1-a_1b_2}, \frac{a_2c_1-a_1c_2}{a_2b_1-a_1b_2}\right) \tag{3}$$

The process of SGTM is as follows:

(1) Locate coordinates of four corners (clockwise from the top left) by using Hough transform to detect lines at the beginning of a go game, Corner[0], Corner[1], Corner[2], Corner[3].

(2) According to the four corner coordinates of the step (1) get the central position of the go image, the midline of the central position. And the intersections of the midline with the upper edge line and the lower edge line of the grid, Mid_line[0], Mid_line[1].

(3) Corner[0], Mid_line[0], Mid_line[1], Corner[3] as a rectangular corner points, in the same way as step (2), get Mid_line1[0], Mid_line1[1]. Obtain the line between point Mid_line1[0] and Mid_line[1], the line between Corner[3], Mid_line[0] and the intersection1 between the two lines, and the intersection2(L[0]), intersection3(L[1]) between the upper edge line and the lower edge line through the intersection1.

(4) Using L[0], Corner[0], Corner[3], L[1] as a rectangular corner points, the midline L2, and the intersections L2[0], L2[1] are obtained between midline L2 with the upper edge line and the lower edge line.

(5) L[0], L[1], L2[1], L2[0] as a rectangular corner points, L[0], Corner[1], Corner[2], L[1] as a rectangular corner point, and Corner[0], L2[0], L2[1], Corner[3] as a rectangular corner points, repeat the step (2)~(4) to get the 8 lines (excluding edge lines) on the left side of midline.

(6) In the same way as step (5), find 8 lines (excluding edge lines) on the right side of the midline, plus 2 edge lines and middle line, and find 19 vertical lines.

(7) For the four corner coordinates (clockwise from Corner[1]) obtained in the step (1), repeat step (2)~(6), identify the 19 horizontal lines; that is, 19 × 19 lines are recognized.

(8) According to 19 × 19 lines, the 361 intersections of go-board are detected.

C. Piece Recognition

Because of the uneven illumination, only color space to locate piece leads to the instability of piece recognition. In this paper, pieces are identified by detecting shape fusion color space. First, Hough transform is applied to detect whether there is a piece at intersection, then calculate the average brightness near the intersection, and use K-Means to classify the piece at the corner into black, white or empty.

Hough transform has good tolerance and robustness for circular detection. Owing to substantial pieces late in game, the standard Hough transform is complex in calculation and has a high demand for memory. The random Hough

Authorized licensed use limited to: Georgia Institute of Technology. Downloaded on September 27,2026 at 21:56:59 UTC from IEEE Xplore. Restrictions apply.

transform (RHT) can effectively improve the problem. In this paper, a new improved RHT in [13] and [14] is adopted: The perpendicular lines of any two chords on the circle are intersected at the center of the circle. Specifically, on the boundary of a circle, select three points, then two points are connected randomly to form two strings and draw the vertical lines of two strings. The intersection of the verticals line is the center of the circle. Figure 4 is the main process of a new improved RHT.

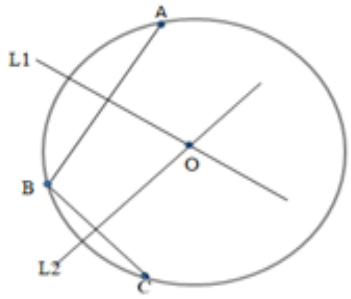


Figure 4. The main process of a new improved RHT.

In figure 4, three points are taken from the edge of the circle: A (x_1, y_1), B (x_2, y_2), C (x_3, y_3), and L_1, L_2 are perpendicular lines of AB, BC, respectively. O (a, b) is the center of the circle.

$$\begin{cases} k_{12} = \dfrac{x_1 - x_2}{y_2 - y_1} \\ k_{23} = \dfrac{x_2 - x_3}{y_3 - y_2} \\ j_{12} = \dfrac{y_1 + y_2 - k_{12}(x_1 + x_2)}{2} \\ j_{23} = \dfrac{y_2 + y_3 - k_{23}(x_2 + x_3)}{2} \\ a = \dfrac{j_{12} - j_{23}}{k_{23} - k_{12}} \\ b = ak_{12} + j_{12} \end{cases} \tag{4}$$

$$r = \sqrt{(x_1 - a)^2 + (y_1 - b)^2} \tag{5}$$

First, the center of the circle is obtained from formula (4), which corresponds to a point in the Hough space, and the peak position of the accumulative array after all possible center coordinates is the center of the target. Then, calculating the mean value of each point to the center of the circle is the radius r by (5).

Color space is an important model for describing images, which often used to describe images as RGB model. However, RGB has no direct semantic interpretation, identify color of piece with HSL color space model. HSL space is made up of hue, saturation, and lightness. Classifying pieces by colors, address it with a simple approach. Compute average color in a 5× 5 window around the intersection. K-Means is used to cluster colors into three clusters: the white, the black and empty.

IV. Experimental Results

A. Dataset

The authors record six complete go games, with 1457 images using different perspectives, various go-boards and uneven illustration. In this dataset, three go-games are in normal light and another three are in various nonuniform illustration. The six games are under normal viewpoints. Furthermore, 100 images of unoccupied board are recorded with different viewpoints. All images in this paper are normalized to 400×360.

B. Result Analysis of Board Recognition

Locating four corners of unoccupied board for the opening of the game by Hough transform to detect lines is the basis of board recognition. 100 images of unoccupied board are based on different view angle from 0 to 90 degrees in the same normal light. The angle is the line of sight from the camera to the board center shown in the left of Figure 5, and the result is shown in the right of the figure. From the figure, it strongly proves that Hough transform can recognize board without pieces successfully when the angle is equal or greater than 40 degrees. It's the essential step for the following SGTM.

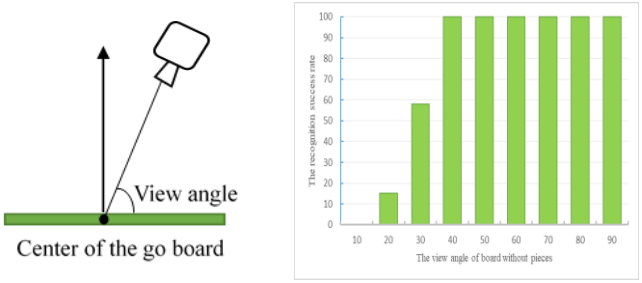


Figure 5. The left is the view angle, the right is the relation between the viewpoints and the recognition.

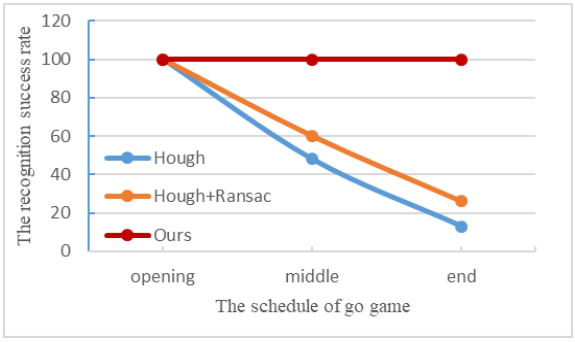


Figure 6. Performance curves evaluated on dataset.

The performance of three approaches of different periods in go game is evaluated on dataset in Figure 6. Each

Authorized licensed use limited to: Georgia Institute of Technology. Downloaded on September 27,2026 at 21:56:59 UTC from IEEE Xplore. Restrictions apply.

of them is described by the recognition success rate on the same normal light, viewpoint and the same go game images. From the figure, in the beginning of the game, three methods can identify board successfully. As the game goes on, the grid lines of board are obscured gradually by substantial pieces. Hough transform lower success rate of detecting lines sharply and Hough transform combined with RANSAC algorithm just improve success rate to some extent. At the end of game, the problem of finding lines can be addressed accurately by the method of this paper.

C. *Result Analysis of Piece Recognition*

For dataset, the performance of different methods is defined by average precision as shown in Table I. Owing to the stochastic algorithm and various result, the program is tested three times. The method of this paper achieves 98.3%, which is better than others. The test dataset contains different viewpoints, uneven illustration. Hough transform is used to detect circle to locate piece, which settle the uneven illustration. Mixed color space, recognize pieces exactly. Figure 7 shows some results on dataset. The first line is the result of board recognition. The second to the last line are the results of pieces recognition. The green is the white piece and the blue represents the black. For example, for the figure, the last two lines, illumination of the image is uneven, some places are brighter than others and the result of piece recognition is that piece is detected where there is no piece in the intersection just using color space [3]. The last line is our method of it.

TABLE I. PERFORMANCE EVALUATED ON DATASET

Algorithm	***Average Precision***
HF + Color [1]	50.2%
HF + RANSAC + Color [2]	63.8%
HF + SGTM + Color [3]	96.9%
Ours [4]	98.3%

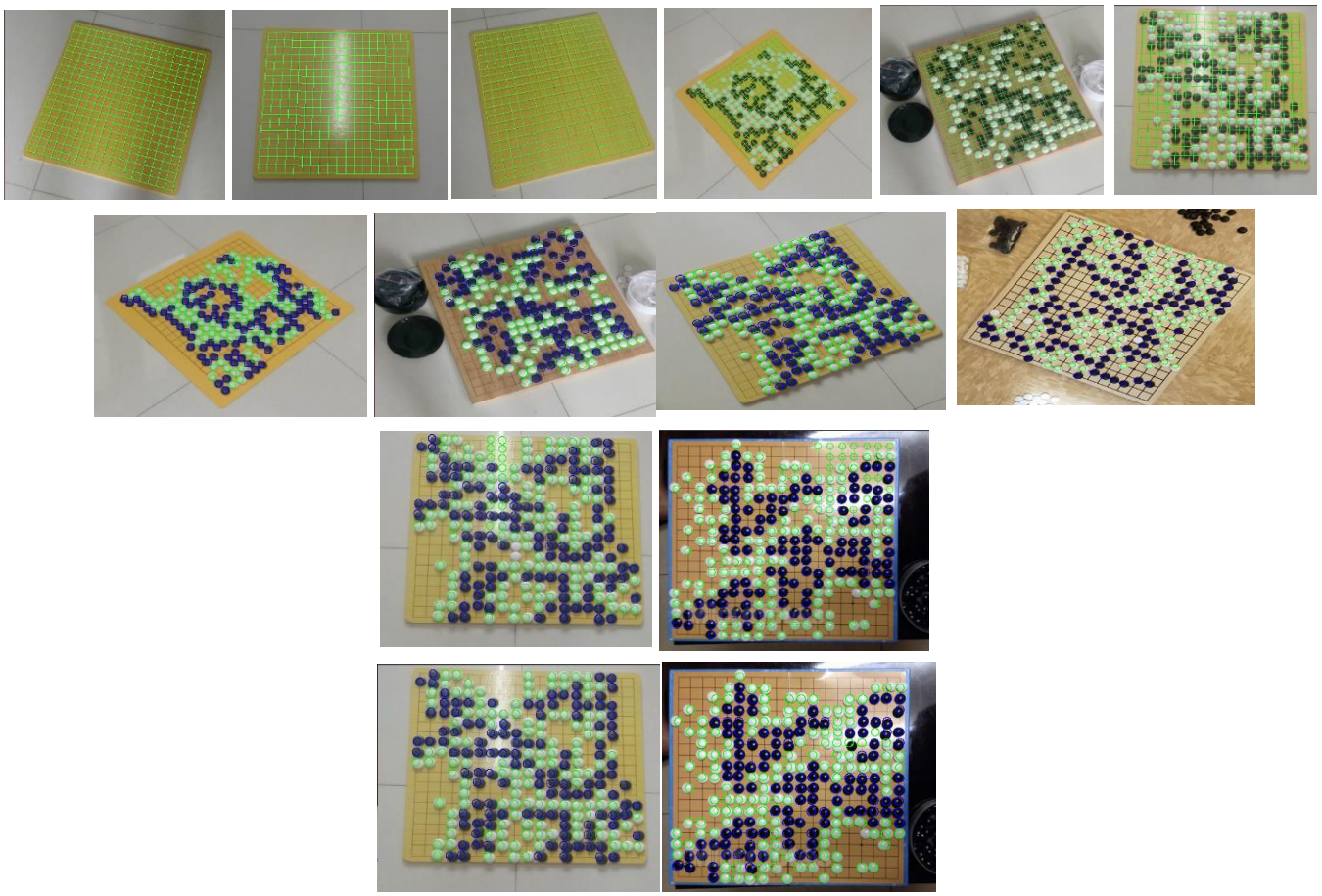


Figure 7. Selected examples of dataset.

Now point out the shortcoming of the method of this paper: low view angle. When the angle is under 10 degrees, the first step, the method of locating four corners by Hough transform to detect lines will fail and so the following steps also fail. Fortunately, corner detection method can be used to locate four corners such as Harris [15].

V. CONCLUSION

In this paper, a method to recognize image in go game that is invariant to occlusions, different viewpoints and illustration variations is presented. Use edge detection to

Authorized licensed use limited to: Georgia Institute of Technology. Downloaded on September 27,2026 at 21:56:59 UTC from IEEE Xplore. Restrictions apply.

preprocess images. Hough transform to detect lines is used to locate four corners for the unoccupied board under the condition that the board is fixed in go game. Then, SGTM is the key of identifying board. The last step, Hough transform to detect circle mixed color space is used to recognize piece. They strongly prove that our method can recognize exactly with large variability, especially in different views and nonuniform illustration.

ACKNOWLEDGMENT

The authors would like to express appreciations to everyone in our laboratory for their helps. This work was supported by the support project of 12th Five-Year Plan of the Ministry of Science and Technology, Intelligent Interactive Integration and Application Demonstration of HuiZhou Culture Communication (2015BAK24B01).

REFERENCES

[1] Kuncheva L I, Constance J H V. Restricted Set Classification with prior probabilities: A case study on chessboard recognition[J]. Pattern Recognition Letters, 2018, 111.J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[2] Xie Y, Tang G, Hoff W A. Geometry-Based Populated Chessboard Recognition[C]// Tenth International Conference on Machine Vision. 2017.

[3] Czyzewski M A. An Extremely Efficient Chess-board Detection for Non-trivial Photos[J]. 2017.

[4] Hirsim T. Extracting Go Game Positions from Photographs[J]. Helsinki University of Technology. 2005.

[5] Nhat V Q, Lee G S. Chessboard and Pieces Detection for Janggi Chess Playing Robot[J]. International Journal of Contents, 2013, 9(4):16-21.

[6] Scher S, Crabb R, Davis J. Making real games virtual: Tracking board game pieces[C]// International Conference on Pattern Recognition. IEEE, 2008:1-4.

[7] Xie Y, Tang G, Hoff W. Chess Piece Recognition Using Oriented Chamfer Matching with a Comparison to CNN[C]// IEEE Winter Conference on Applications of Computer Vision. IEEE Computer Society, 2018:2001-2009.

[8] Shiba K, Mori K. Detection of Go-board contour in real image using genetic algorithm[C]// Sice 2004 Conference. IEEE, 2005:2754-2759 vol. 3.

[9] Jialin Ding, ChessVision: Chess Board and Piece Recognition, 2015. Stanford University, USA.

[10] Seewald A K. Automatic extraction of go game positions from images: A multi-strategical approach to constrained multi-object recognition[M].Taylor & Francis,Inc.2010.

[11] Srisuphab A, Silapachote P, Chaivanichanan T, et al. An application for the game of Go: Automatic live Go recording and searchable Go database[J]. 2012:1-6.

[12] Duda R O. Use of the Hough transformation to detect lines and curves in pictures[J]. Cacm, 1972, 15(1):11-15.

[13] Yue J, Xiang X Z. An improved algorithm of Hough circle detection[J]. Applied Science & Technology, 2006.

[14] Wang L, Chen L Q. Concentric circle detection based on chord mid-point Hough transform[J]. Journal of Computer Applications, 2009, 29(7):1937-1939.

[15] Rosten E, Drummond T. Machine learning for high-speed corner detection[C]// European Conference on Computer Vision. Springer, Berlin, Heidelberg, 2006:430-44

Authorized licensed use limited to: Georgia Institute of Technology. Downloaded on September 27,2026 at 21:56:59 UTC from IEEE Xplore. Restrictions apply.